Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

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Degenerado

Maximize

$$Z = 2.00x_1 + 1.00x_2$$

Subject to

$$3.00x_1 + 1.00x_2 \leq 6.00$$
$$1.00x_1 - 1.00x_2 \leq 2.00$$
$$0.00x_1 + 1.00x_2 \leq 3.00$$

Simplex Table

Z	x_1	x_2	s_1	s_2	s_3	b
1	-2	-1	0	0	0	0
0	3	1	1	0	0	6
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Intermediate Tables

Pivoting 1

Most Negative

Column 2 (-2.00)

Z	x_1	x_2	s_1	s_2	s_3	b
1	-2	-1	0	0	0	0
0	3	1	1	0	0	6
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Fractions

Z	x_1	x_2	s_1	s_2	s_3	b	Frac
1	-2	-1	0	0	0	0	
0	3	1	1	0	0	6	2.00
0	1	-1	0	1	0	2	2.00
0	0	1	0	0	1	3	inf

$6.00/3.00 = 2.00$
$2.00/1.00 = 2.00$
$3.00/0.00 = inf$
Smallest fraction: 2.00 $\rightarrow$ Pivot: row 3

Pivot

Z	x_1	x_2	s_1	s_2	s_3	b
1	-2	-1	0	0	0	0
0	3	1	1	0	0	6
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Canonization

$R_3 \leftarrow R_3/1.00$

Z	x_1	x_2	s_1	s_2	s_3	b
1	-2	-1	0	0	0	0
0	3	1	1	0	0	6
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

$R_1 \leftarrow R_1 + 2.00R_3$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	3	1	1	0	0	6
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

$R_2 \leftarrow R_2 + -3.00R_3$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	0	4	1	-3	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

$R_4 \leftarrow R_4 + -0.00R_3$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	0	4	1	-3	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Pivot Result

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	0	4	1	-3	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Degenerate Base

The variable s_1 is part of the base but has a value of 0. Therefore, this is a degenerate Basic Feasible Solution (BFS).

Pivoting 2

Most Negative

Column 3 (-3.00)

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	0	4	1	-3	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Fractions

Z	x_1	x_2	s_1	s_2	s_3	b	Frac
1	0	-3	0	2	0	4	
0	0	4	1	-3	0	0	0.00
0	1	-1	0	1	0	2	-2.00
0	0	1	0	0	1	3	3.00

$0.00/4.00 = 0.00$
$2.00/-1.00 = -2.00$
$3.00/1.00 = 3.00$
Smallest fraction: 0.00 $\rightarrow$ Pivot: row 2

Pivot

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	0	4	1	-3	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

Canonization

$R_2 \leftarrow R_2/4.00$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	-3	0	2	0	4
0	0	1	0.25	-0.75	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

$R_1 \leftarrow R_1 + 3.00R_2$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	-1	0	1	0	2
0	0	1	0	0	1	3

$R_3 \leftarrow R_3 + 1.00R_2$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	1	0	0	1	3

$R_4 \leftarrow R_4 + -1.00R_2$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	0	-0.25	0.75	1	3

Pivot Result

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	0	-0.25	0.75	1	3

Degenerate Base

The variable x_2 is part of the base but has a value of 0. Therefore, this is a degenerate Basic Feasible Solution (BFS).

Pivoting 3

Most Negative

Column 5 (-0.25)

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	0	-0.25	0.75	1	3

Fractions

Z	x_1	x_2	s_1	s_2	s_3	b	Frac
1	0	0	0.75	-0.25	0	4	
0	0	1	0.25	-0.75	0	0	-0.00
0	1	0	0.25	0.25	0	2	8.00
0	0	0	-0.25	0.75	1	3	4.00

$0.00/-0.75 = -0.00$
$2.00/0.25 = 8.00$
$3.00/0.75 = 4.00$
Smallest fraction: 4.00 $\rightarrow$ Pivot: row 4

Pivot

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	0	-0.25	0.75	1	3

Canonization

$R_4 \leftarrow R_4/0.75$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.75	-0.25	0	4
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	0	-0.333	1	1.33	4

$R_1 \leftarrow R_1 + 0.25R_4$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.667	0	0.333	5
0	0	1	0.25	-0.75	0	0
0	1	0	0.25	0.25	0	2
0	0	0	-0.333	1	1.33	4

$R_2 \leftarrow R_2 + 0.75R_4$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.667	0	0.333	5
0	0	1	0	0	1	3
0	1	0	0.25	0.25	0	2
0	0	0	-0.333	1	1.33	4

$R_3 \leftarrow R_3 + -0.25R_4$

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.667	0	0.333	5
0	0	1	0	0	1	3
0	1	0	0.333	0	-0.333	1
0	0	0	-0.333	1	1.33	4

Pivot Result

Z	x_1	x_2	s_1	s_2	s_3	b
1	0	0	0.667	0	0.333	5
0	0	1	0	0	1	3
0	1	0	0.333	0	-0.333	1
0	0	0	-0.333	1	1.33	4

Results

	Z	x_1	x_2	s_1	s_2	s_3	b
	1	0	0	0.667	0	0.333	5
	0	0	1	0	0	1	3
	0	1	0	0.333	0	-0.333	1
	0	0	0	-0.333	1	1.33	4

Objective Value

Z = 5.00

Variables

$x_1 = 1.00$
$x_2 = 3.00$

Slack or Surplus

$s_1 = 0.00$
$s_2 = 4.00$
$s_3 = 0.00$